

Costume Recognition System for a Robot

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Motivation



- › DiMi
- › Visual recognition for a humanoid robot
- › Yearly attendance at Fantasy Basel



Project Idea



"Spider Man"

Project Idea

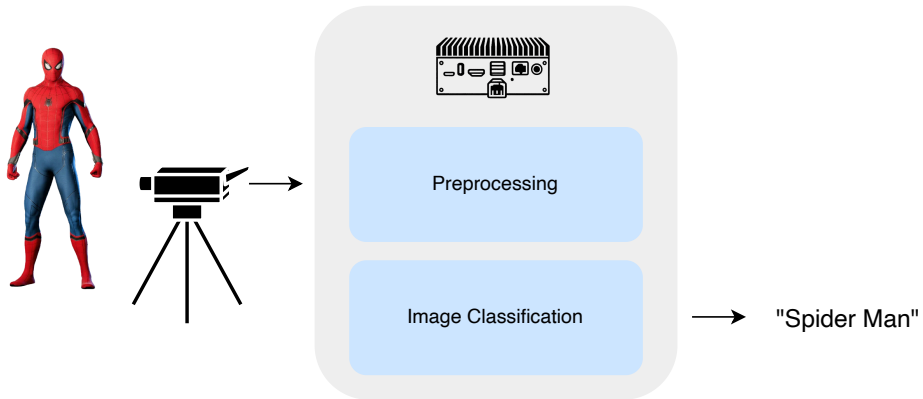
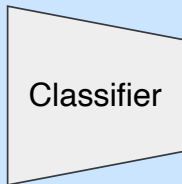
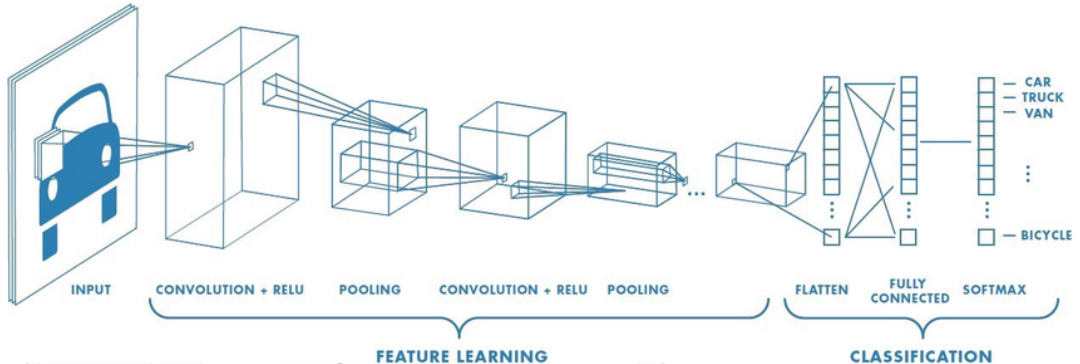


Image Classification



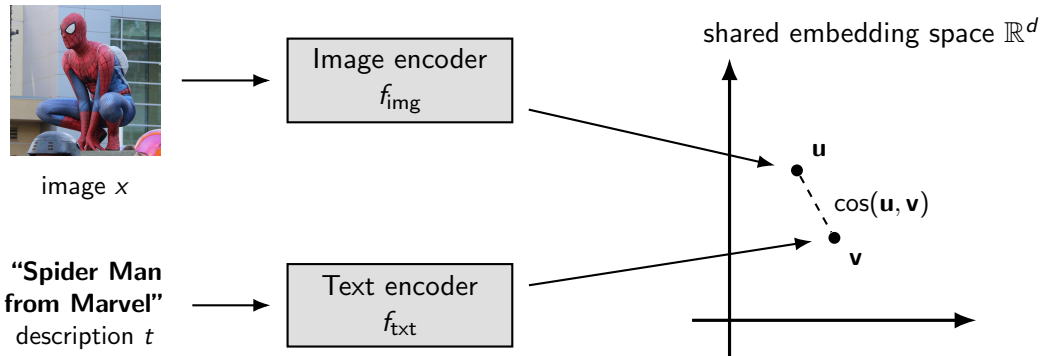
Spider Man

Approach 1: Convolutional Neural Network (CNN)



Source: Saha, "A Comprehensive Guide to CNNs", towardsdatascience.com

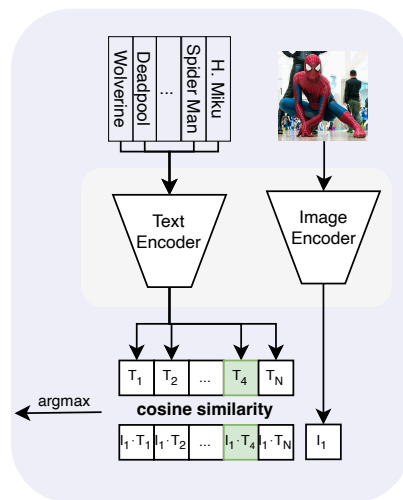
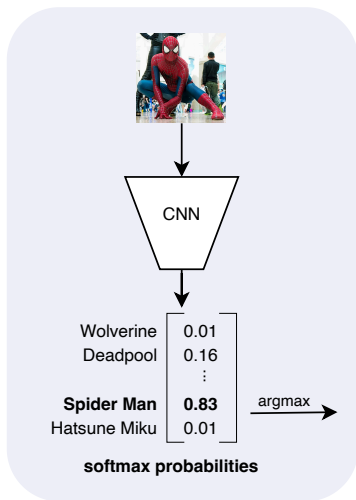
Approach 2: Vision Language Model



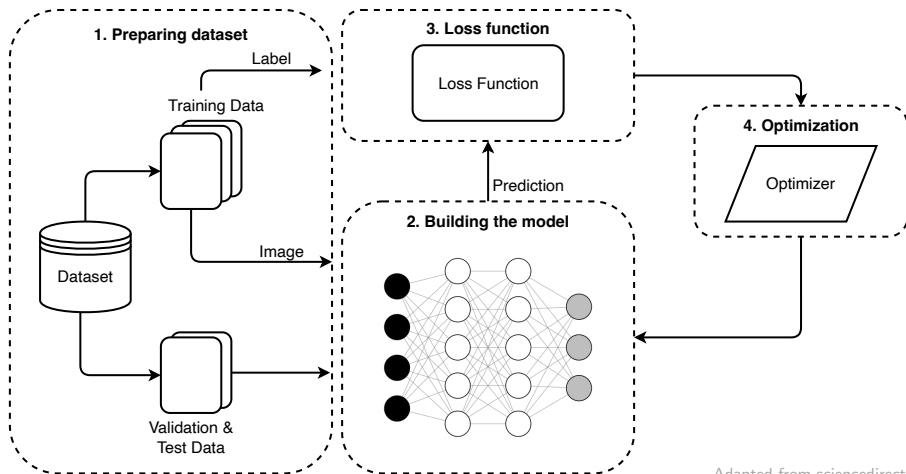
Discriminative

vs.

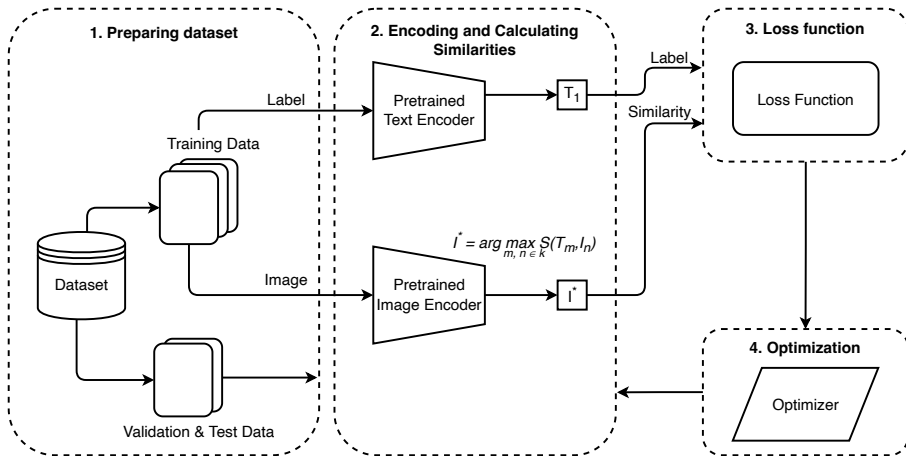
Similarity-Based



Training a neural network



Finetuning a vision language model



What kind of data do we need ?



Spiderman Costumes

Source: Wikimedia Commons



Spiderman Costumes

Source: Wikimedia Commons



Deadpool

Source: Marvel Wiki (Fandom)



Wolverine Cosplays

Source: Wikimedia Commons

Created Dataset

30 Classes

Sources: Wikimedia Commons, Reddit

	Images	Train (70%)	Val (15%)	Test (15%)
Cosplay Images	5252	3669	779	804

Training the neural network - ResNet-18

Hyperparameter Sweep

Parameter	Range
Learning rate	$[3 \times 10^{-4}, 3 \times 10^{-3}]$
Weight decay	$\{10^{-4}, 10^{-3}\}$
Dropout p	$\{0.0, 0.2, 0.4\}$
Label smoothing	$\{0.05, 0.1, 0.15, 0.2\}$
CutMix α	$\{0, 0.2, 0.4, 0.6\}$
Jitter strength	$\{\text{mild}, \text{strong}\}$
Rotation degrees	$\{0, 10, 20\}$
Random erasing	$\{\text{true}, \text{false}\}$

Candidate Configurations

Parameter	fiery-6	treasured-22	devoted-19	grateful-11
lr	4.42e-4	4.67e-4	1.07e-3	4.63e-4
weight_decay	0.001	0.001	0.001	0.0001
dropout_p	0.2	0.0	0.2	0.2
label_smoothing	0.2	0.2	0.2	0.2
cutmix_alpha	0.6	0.6	0.6	0.6
jitter_strength	strong	strong	strong	strong
rotation_degrees	0	0	10	10
aug_erasing	false	true	false	true

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Jitter strength	$\{\text{mild}, \text{strong}\}$
Rotation degrees	$\{0, 10, 20\}$
Random erasing	$\{0, 1\}$

Approach 1: ResNet-18 Results

Configuration	From Scratch	Pretrained
	Val %	Val %
fiery-6	81.9	93.5
treasured-22	80.1	92.4
devoted-19	80.9	91.4
grateful-11	81.4	91.3

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Finetuning the Embedding Model

Model Selection: xlm-roberta-base-ViT-B-32 (CLIP) or ViT-B-16-SigLip2 (SigLIP2)

Model	Latency Mean / Median	Throughput
CLIP	13.7 ms / 13.6 ms	119 img/s
SigLIP2	48.8 ms / 47.6 ms	22 img/s

Table: Inference cost

Dim	Top-1	Top-5
512	61.6%	81.0%
768	84.5%	93.8%

Table: Zero-shot evaluation

Finetuning the Embedding Model

Caption Design: "*character* from *franchise*" vs.
"A person dressed up as *character* from *franchise*"

Finetuning the Embedding Model

Finetuning with Batch Size 32, Learning Rate 10^{-6} , Weight Decay 0.1

Checkpoint	Val Top-1	Val Top-5
Pretrained baseline	84.5%	93.8%
Epoch 1	96.0%	99.9%
Epoch 2	96.9%	100.0%
Epoch 3	97.7%	100.0%
Epoch 4	97.8%	100.0%
Epoch 5	97.9%	100.0%

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Comparison of Approaches

Table: Top-1 Accuracy of the final models

Model	Test %
ResNet-18, from scratch	78.1
ResNet-18, pretrained	90.7
SigLIP2, zero-shot	87.6
SigLIP2, finetuned	97.4

Comparison of Approaches

Table: Top-1 Accuracy of the final models

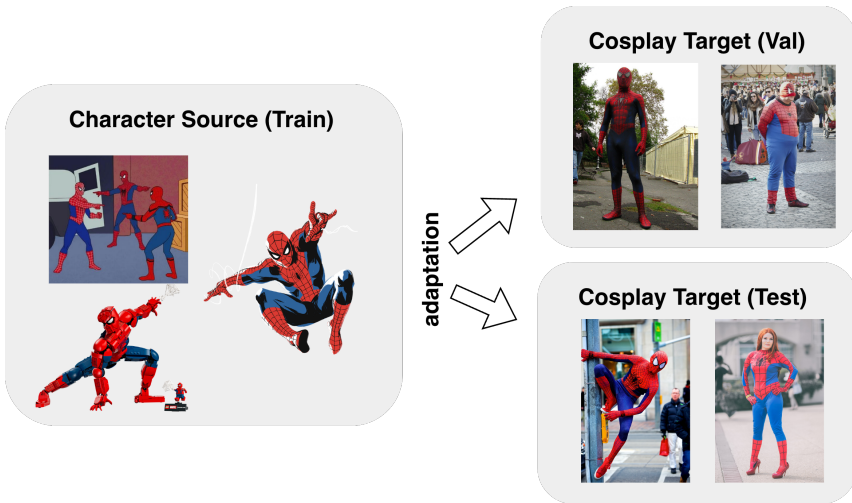
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Domain Adaptation



Updated Dataset

	Classes	Images	Train	Val	Test
Cosplay Images	30	5252	3669	779	804
Subset of Character images	10		1626		
New Character Images	10	2610	1934	341	335

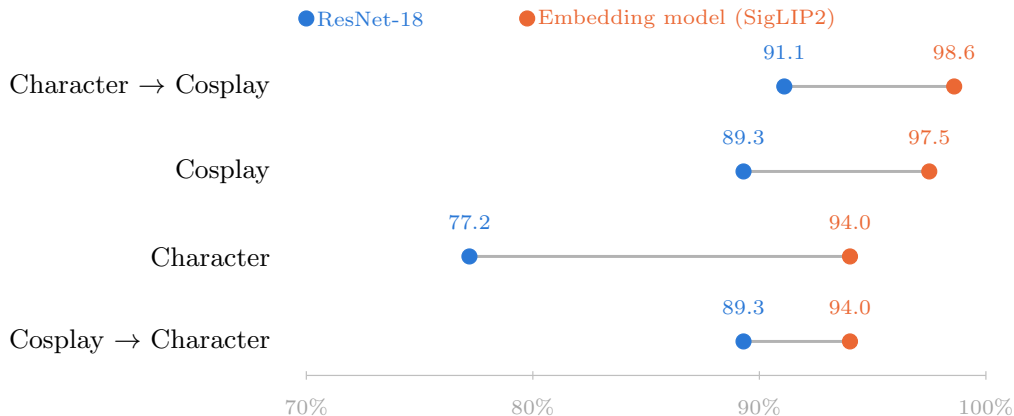
Domain Adaptation Experiments

- › How feasible is training on Character Images compared to Cosplay Images ?

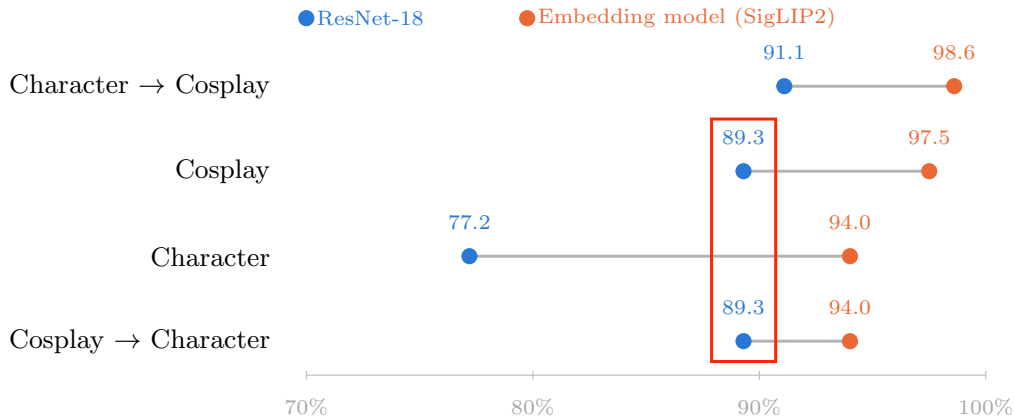
Training condition	First domain	Second domain
Cosplay	Cosplay	—
Character	Character	—
Character → Cosplay	Character	Cosplay
Cosplay → Character	Cosplay	Character

- › Extensibility: How easily can the dataset be extended ?

Results of Domain Adaptation Experiments

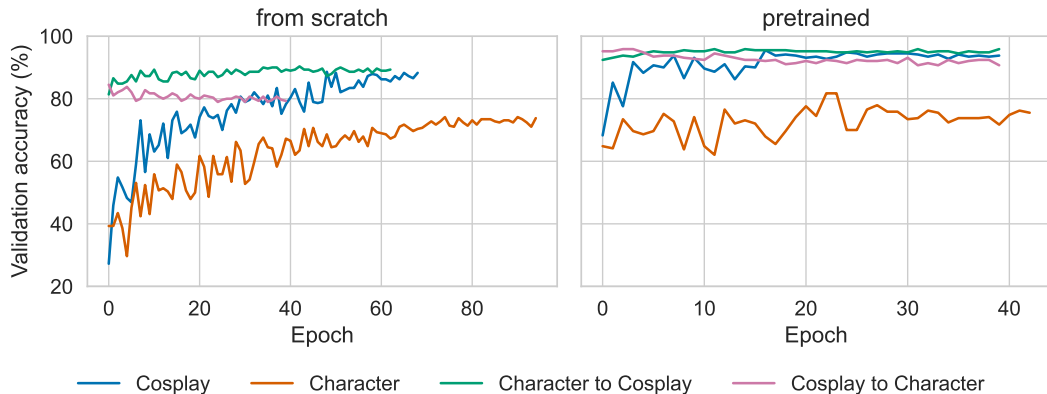


Results of Domain Adaptation Experiments

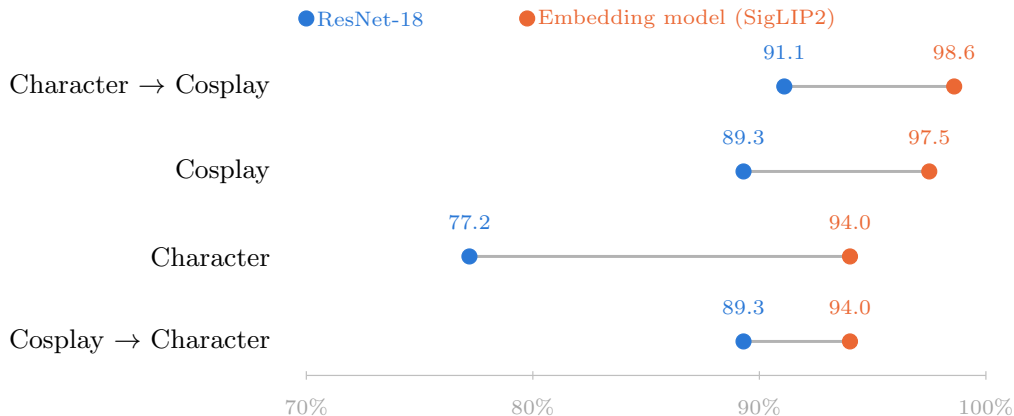


Results of Domain Adaptation Experiments

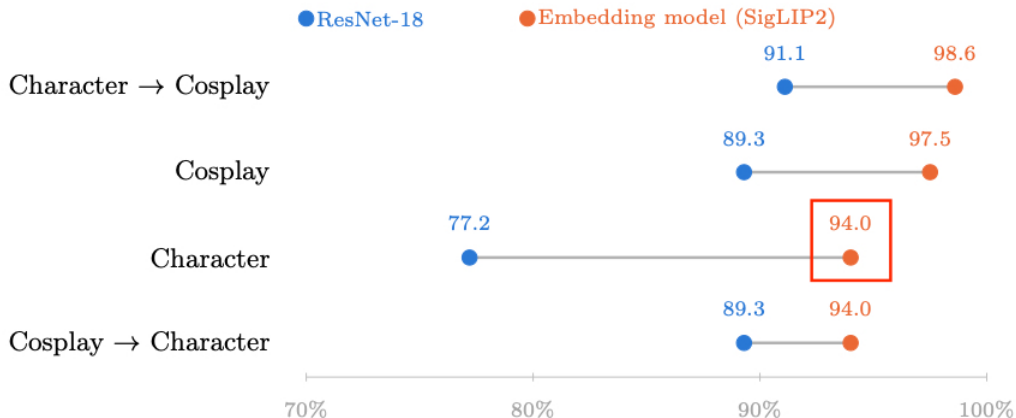
ResNet-Training



Results of Domain Adaptation Experiments



Results of Domain Adaptation Experiments

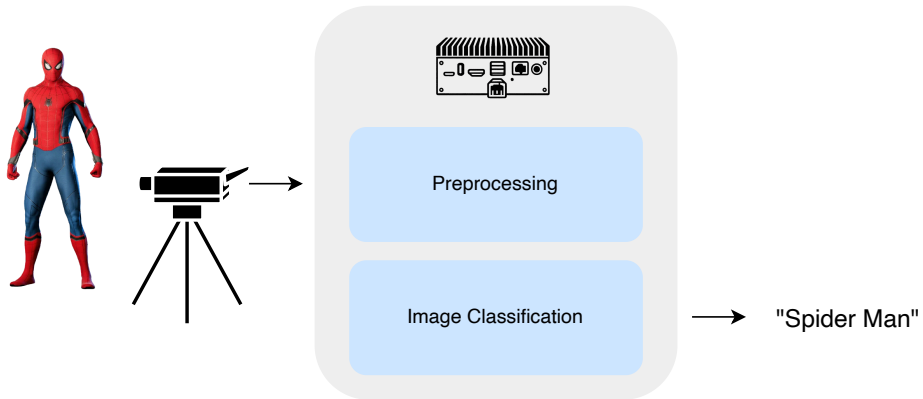


Extensibility

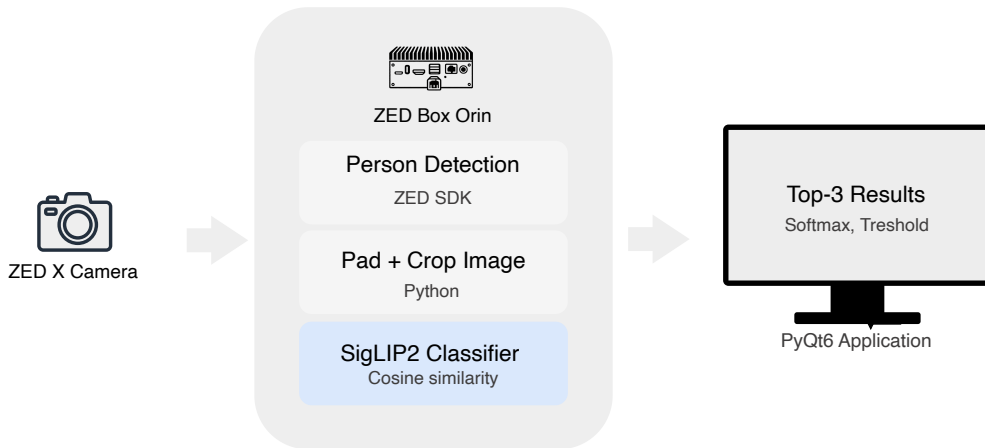
- › Training on Character Images is feasible for Embedding Model
- › 30 Cosplay Classes + 10 Character Classes

Final model with 40 classes and 96.2% Top-1 Test Accuracy

System Overview



System Overview



Demo

Conclusion

- › Successful Prototype with 40 recognized characters

Limitations:

- › Dataset
- › Experiments

Future Work:

- › Extend the dataset
- › Further develop embedding model
- › Test the system at the next Fantasy Basel

Questions ?

Image Sources

Slide 1 Spiderman: <https://www.freepnglogos.com/images/spiderman-10243.html>

Robot: PAL Robotics / ROS Robots — <https://robots.ros.org/tiago++/>

Slide 2 Spider-Man: https://marvels-spider-man.fandom.com/wiki/Stark_Suit

Robot: PAL Robotics — <https://pal-robotics.com/>

ZED BOX: Generated with ChatGPT

Slides 3, 5, 6 Spider-Man: Wikimedia Commons — <https://commons.wikimedia.org/>

Slide 19 Spider-Man: Marvel, <https://beyondamazingexhibition.com/>

Spider-Man: The Bradley Scout, *Rumored Three Spider-Mans in One* —

<https://www.bradleyscout.com/voice/rumored-three-spider-mans-in-one/>

Lego Spider-Man: <https://www.lego.com/en-ch/product/spider-man-hero-figure-76346>

Additional images: Wikimedia Commons — <https://commons.wikimedia.org/>